OFFICIAL COLLEGE PROJECT ABSTRACT · PROJECT 21 OF 55

Credit Card Fraud Detection using Machine Learning Algorithms

Domain: Machine Learning & Financial Risk | **Module:** project_21_credit_card_fraud

OFFICIAL ABSTRACT

This Project is focused on credit card fraud detection in real world scenarios. Nowadays credit card frauds are drastically increasing in number as compared to earlier times. Criminals are using fake identity and various technologies to trap the users and get the money out of them. Therefore, it is very essential to find a solution to these types of frauds. In this proposed project we designed a model to detect the fraud activity in credit card transactions. This system can provide most of the important features required to detect illegal and illicit transactions. As technology changes constantly, it is becoming difficult to track the behavior and pattern of criminal transactions. To come up with the solution one can make use of technologies with the increase of machine learning, artificial intelligence and other relevant fields of information technology; it becomes feasible to automate this process and to save some of the intensive amounts of labor that is put into detecting credit card fraud. Initially, we will collect the credit card usage data-set by users and classify it as trained and testing dataset using a LSTM algorithm and decision trees. Using this feasible algorithm, we can analyze the larger data-set and user provided current data-set. Then augment the accuracy of the result data. Proceeded with the application of processing of some of the attributes provided which can find affected fraud detection in viewing the graphical model of data visualization. The performance of the techniques is gauged based on accuracy, sensitivity, and specificity, precision.

📌 Project Summary & System Specifications

Project Title	Credit Card Fraud Detection using Machine Learning Algorithms
Specialization / Category	Machine Learning & Financial Risk

Target Technology Stack	Python (Flask), PyTorch / TensorFlow / Scikit-Learn, Modern Web UI
Default Backend Port	`:5021`
Local Implementation	`project_21_credit_card_fraud`